Chapter 6

Microbial Nutrition and Growth

EPAWX4



Mold is a type of fungi that grows on food and other organic matters. It breaks down the complex substances into simpler ones and extracts nutrient for its growth from them.

Chapter Outline

6.1 Microbial Nutrients

6.2 Nutrient Requirement of Microorganisms

6.3 Nutritional Types of Microorganisms

6.4 Photosynthesis

6.5 Microbial Growth

6.6 Measurement of Microbial Growth

Learning Objectives

After studying this chapter the student will be able,

- *To know the essential nutrients required by bacterial cell.*
- *To differentiate between macronutrients and micronutrients.*
- *To describe an organism based on the sources of carbon and energy.*
- *To compare the photosynthesis process in plant, algae and bacteria.*
- *To understand the phases of growth in bacterial growth curve.*
- *To know the methods of counting bacteria.*

6.1 Microbial Nutrition

All living organisms on this planet require energy for the normal functioning, growth and reproduction. Likewise, microorganisms acquire energy from various organic and inorganic compounds, light and CO_2. The requirement of energy depends on their need and metabolic ability.

6.2 Nutrient Requirement of Microorganisms

Microorganisms requires macronutrients, micronutrients and growth factors, for their growth. These nutrients help in constructing the cellular components like proteins, nucleic acids and lipids.

Macronutrients

Elements that are required in large amounts are called macronutrients. Nitrogen (N), Carbon (C), Oxygen (O), Hydrogen (H), Sulphur (S) and Phosphorus (P), Potassium (K), Calcium (Ca), Magnesium (Mg) and Iron (Fe) are macroelements.